

07/10/20

① Esercizio 1

(a) $V = V(x)$? $E_x = - \frac{\partial V}{\partial x}$

$x \leq -d$: $V(-d) = 0 \Rightarrow V(x) = 0$

$-d \leq x \leq 0$: $E(x) = \frac{E_0}{d} x + E_0$

$$V(x) = V(-d) - \int_{-d}^x E(x') dx'$$

$$V(x) = - \int_{-d}^x \left(\frac{E_0}{d} x' + E_0 \right) dx' = - \left\{ \frac{E_0}{d} \left[\frac{x'^2}{2} \right]_{-d}^x + E_0 [x']_{-d}^x \right\}$$

$$= -E_0 \left(\frac{x^2}{2d} - \frac{d^2}{2d} + x + d \right) = -E_0 \left(\frac{x^2}{2d} + x + \frac{d}{2} \right)$$

$$\underline{0 \leq x \leq d} : E(x) = -\frac{E_0}{d} x + E_0$$

$$V(x) = \underbrace{V(0)}_{\downarrow -E_0 \frac{d}{2}} - \int_0^x E(x') dx'$$

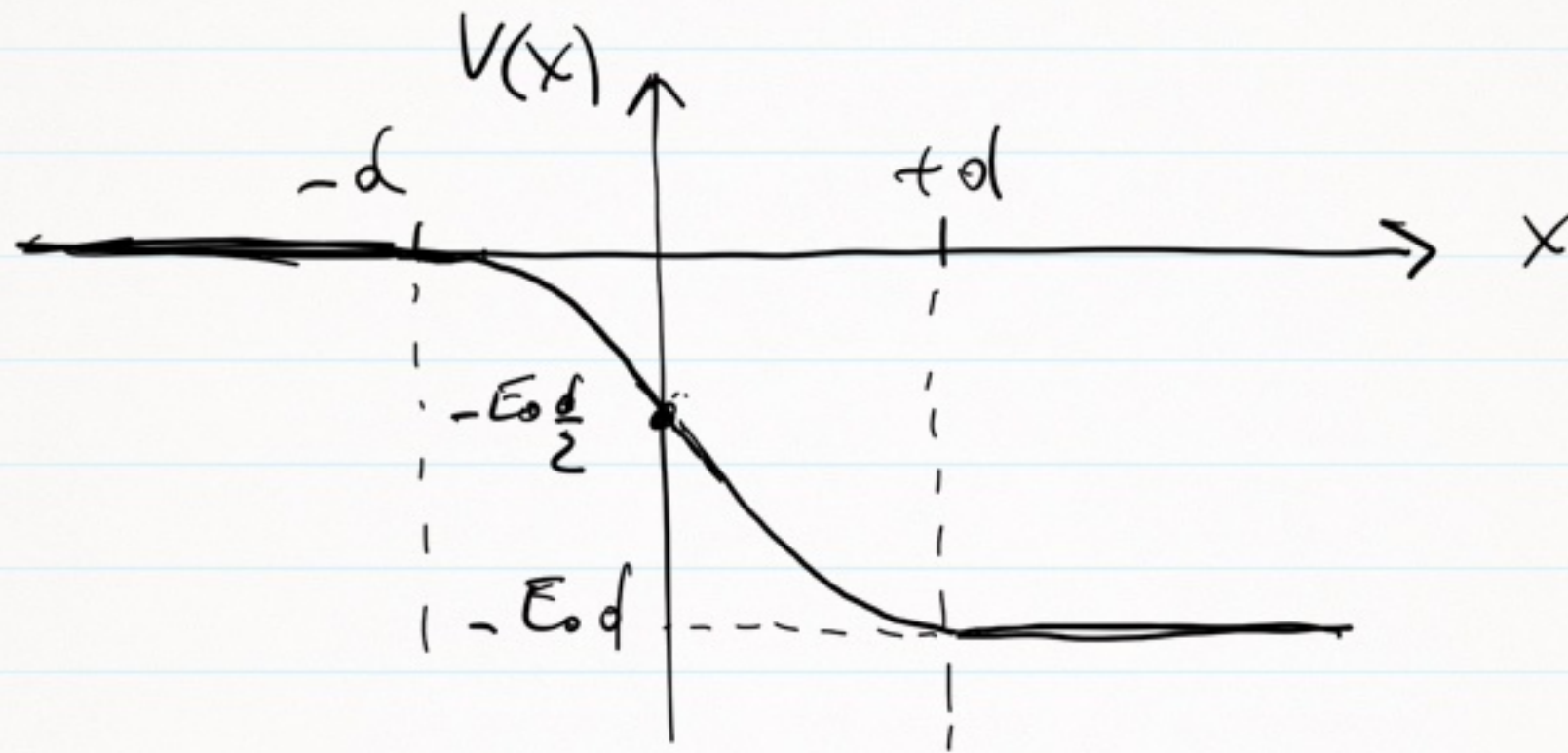
$$V(x) = -E_0 \cdot \frac{d}{2} + \int_0^x \left(\frac{E_0}{d} x' - E_0 \right) dx'$$

$$= -E_0 \frac{d}{2} + \frac{E_0}{d} \left[\frac{x'^2}{2} \right]_0^x - E_0 [x']_0^x$$

$$= -E_0 \cdot \frac{d}{2} + \frac{E_0}{d} \frac{x^2}{2} - E_0 x$$

$$= E_0 \left(\frac{x^2}{2d} - x - \frac{d}{2} \right) \rightarrow @ x=d, V(d) = -E_0 d$$

$$\underline{x \geq d} : V(x) = -E_0 \cdot d$$



(b) $e = -1.6 \cdot 10^{-19} \text{ C}$

$x < -d$: INITIAL $E_i = E_{K,i} + U_i = \frac{1}{2} m_e v_0^2 + \cancel{e V_i}$

$x > d$: FINAL $E_f = E_{K,f} + U_f = 0 + e V_f$

$\boxed{E_i < E_f} \Rightarrow E_{K,i} < e V_f = e \Delta V = -e \overset{-E_0 d}{E_0 d}$

$$E_p > - \frac{E_{K,i}}{e d} = - \frac{m_e v_0^2}{2 e d} = \frac{9.11 \cdot 10^{-31} \cdot 10^{12}}{2 \cdot 1.6 \cdot 10^{-19} \cdot 2 \cdot 10^{-2}}$$

$$\approx 1.4 \cdot 10^2 \frac{V}{m}$$

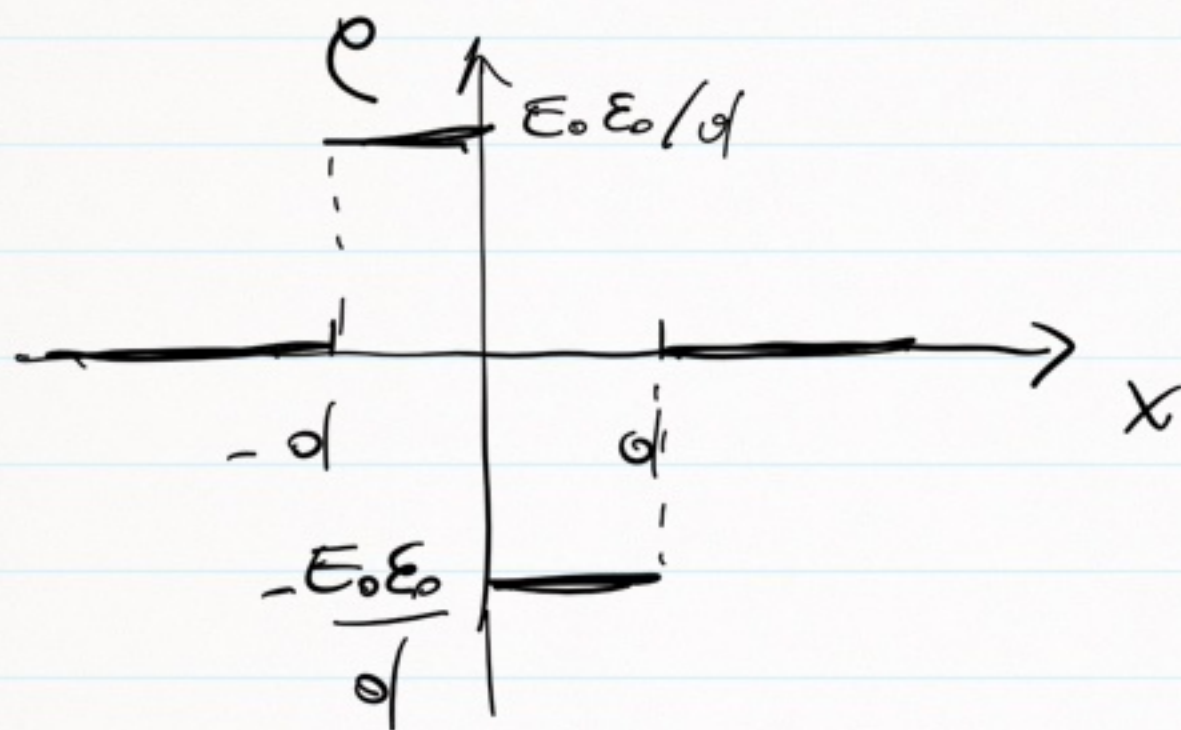
$$(c) \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Rightarrow \frac{dE_x}{dx} = \frac{\rho}{\epsilon_0} \Rightarrow \rho = \epsilon_0 \cdot \frac{dE_x}{dx}$$

• $x < -d \vee x > d$: $\rho = 0 \text{ C/m}^3$

• $-d < x < 0$: $\rho = \epsilon_0 \frac{E_0}{d} = \frac{8.85 \cdot 10^{-12} \cdot 1.4 \cdot 10^2}{2 \cdot 10^{-2}}$

$$\approx 6.2 \cdot 10^{-8} \text{ C/m}^3$$

• $0 < x < d$: $\rho = -\epsilon_0 \frac{E_0}{d} \approx -6.2 \cdot 10^{-8} \text{ C/m}^3$



② Esercizio 2 : $V = ax^2 + bxy - cy^2$

potenziale?

1 calcolo $\vec{E} = -\vec{\nabla} V$: $E_x = -\frac{\partial V}{\partial x} = -2ax - by$

$$E_y = -\frac{\partial V}{\partial y} = -bx + 2cy$$

2 campo irrotazionale? $\vec{\nabla} \times \vec{E} \stackrel{?}{=} 0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix} = \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{k} = (-b + b) \hat{k} \\ \stackrel{!}{=} 0 \hat{k}$$

Sì, potenziale!

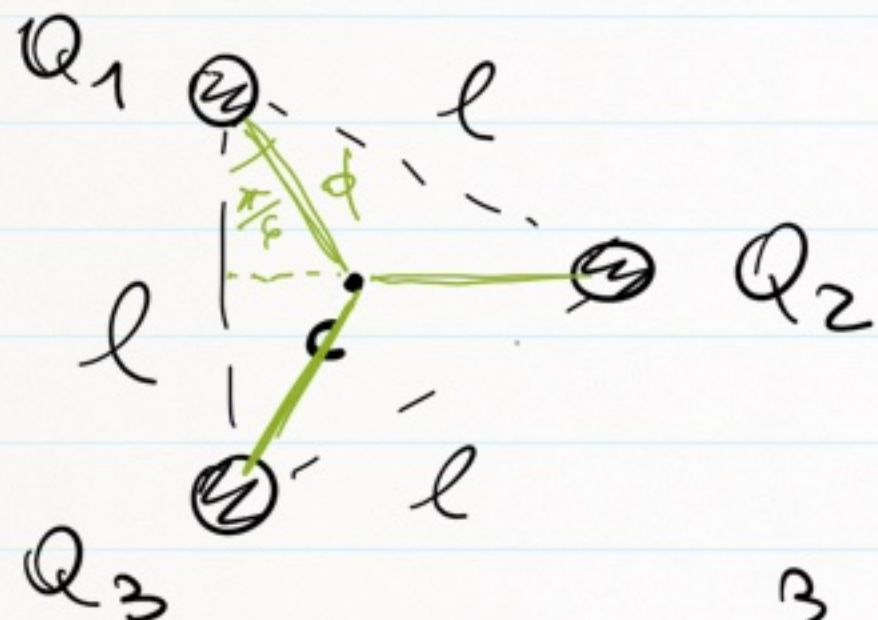
$$\rho = ? \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = -2a + 2a = 0$$

③ Esercizio 3

$$V_c = ?$$

$$C \equiv (x_c, y_c, z_c)$$



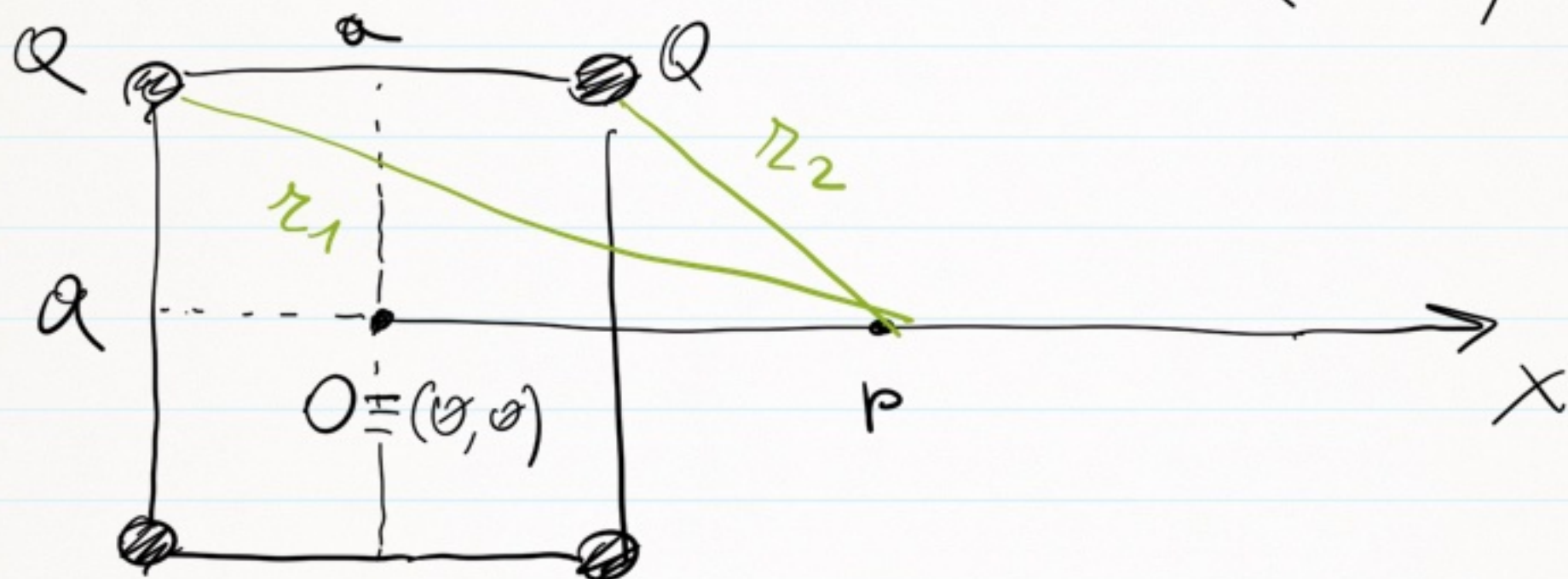
$$d = \frac{l/2}{\cos \pi/6} = \frac{l}{\sqrt{3}}$$

$$V_c = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^3 \frac{Q_i}{\sqrt{(x_c - x_i)^2 + (y_c - y_i)^2 + (z_c - z_i)^2}}$$

$$V_c = \frac{\sqrt{3}}{4\pi\epsilon_0 l} (Q_1 + Q_2 + Q_3) = \text{e.g.} \begin{cases} \frac{3\sqrt{3}}{4\pi\epsilon_0 l} Q, & Q_1 = Q_2 = Q_3 \\ 0, & Q_1 + Q_2 + Q_3 = 0 \end{cases}$$

④ Esercizio 5

$$P \equiv (x, \theta)$$



$$V(x) = 2 \cdot \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$= \frac{Q}{2\pi\epsilon_0} \left(\frac{1}{\sqrt{(x+a/2)^2 + (a/2)^2}} + \frac{1}{\sqrt{(x-a/2)^2 + (a/2)^2}} \right)$$

POSIZIONE INIZIALE : $E_i = E_{K,i} + U_i = \frac{1}{2} m_0 v^2$

$$Q_0 V_i$$

$$V(\infty) = 0$$

PUNTO

FINALE ? $E_f = E_{K,f} + U_f$

Analisi $V(x)$ necessaria per capire ...

$$\frac{dV(x)}{dx} = 0$$

$$- \frac{Q}{4\pi\epsilon_0} \left\{ \left[(x+a/2)^2 + (a/2)^2 \right]^{-3/2} \cdot (x+a/2) + \left[(x-a/2)^2 + (a/2)^2 \right]^{-3/2} \cdot (x-a/2) \right\} = 0$$

$$\begin{aligned} & [(x+a/2)^2 + (a/2)^2]^{-3/2} \cdot (x+a/2) = \\ & = [(x-a/2)^2 + (a/2)^2]^{-3/2} \cdot \left(\frac{a}{2} - x\right) \end{aligned}$$

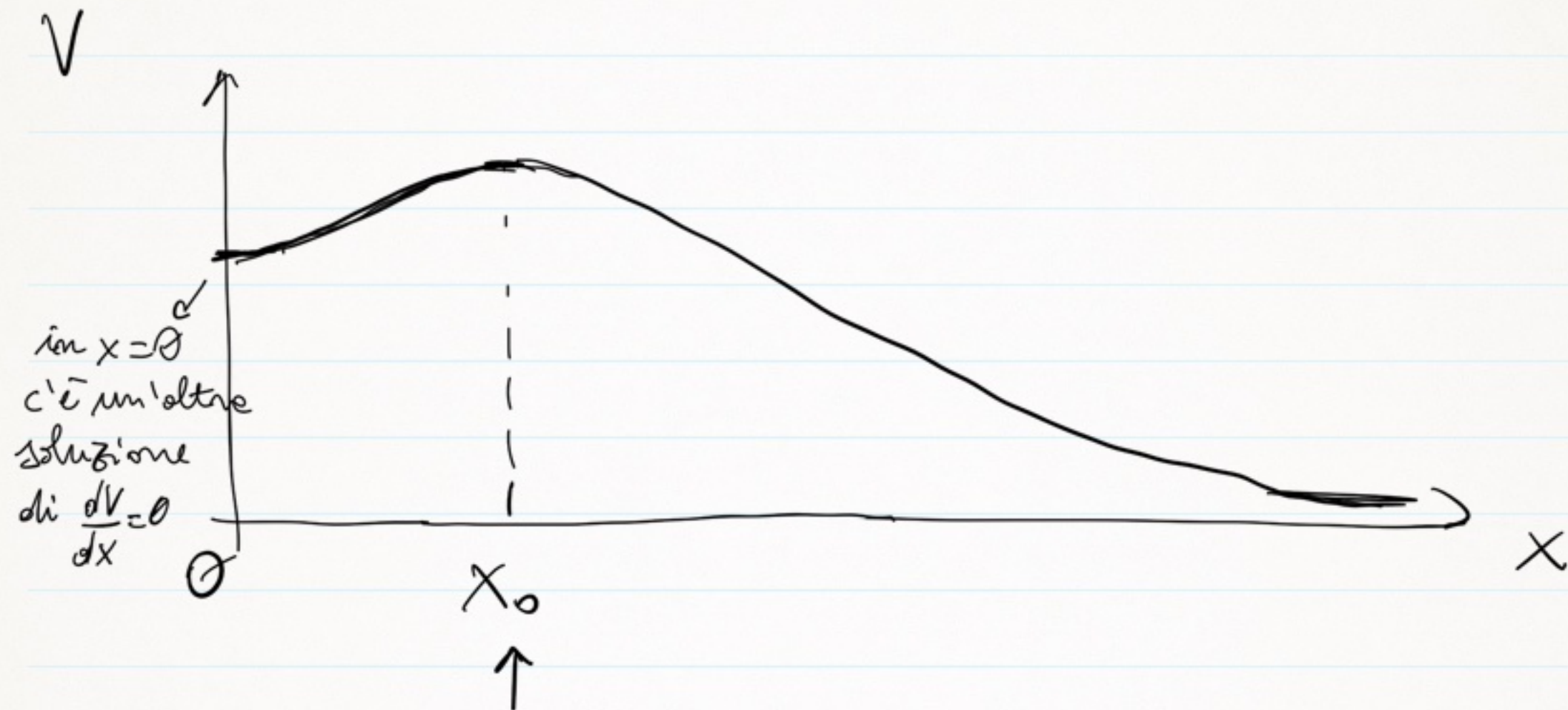
$$\frac{a/2 + x}{a/2 - x} = \left[\frac{(x+a/2)^2 + (a/2)^2}{(x-a/2)^2 + (a/2)^2} \right]^{3/2} \quad \leftarrow$$

per $x > 0$, $x_0 \simeq 0.38674 \text{ cm}^*$

$$V(x=0) = \frac{Q\sqrt{2}}{4\pi\epsilon_0 a} = \frac{10^{-6}\sqrt{2}}{4\pi \cdot 8.85 \cdot 10^{-12}} \stackrel{(?)}{=} 0.05 \cdot 10^6 \text{ V}$$

$$V(x=x_0) = 5.27 \cdot 10^6 \text{ V}$$

[* potete verificare il risultato ^{numerico} V_{approx} con EQUATION SOLVER
 o il WOLFRAM disponibile su internet (free)]

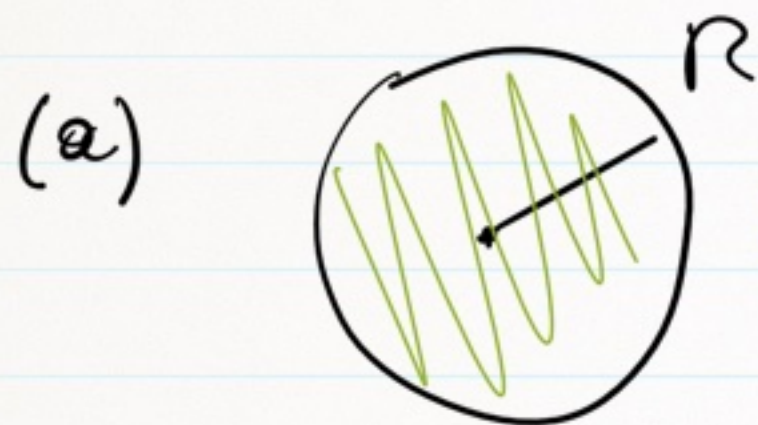


$$\Rightarrow E_i = E_f \Rightarrow E_{K,i} = U_f$$

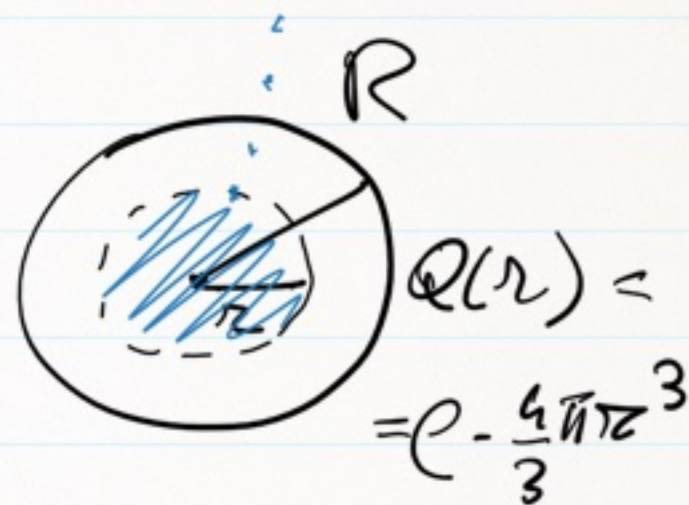
$$\frac{1}{2} m_0 v^2 = Q_0 \cdot V(x_0)$$

$$v = \sqrt{\frac{2 Q_0 V(x_0)}{m_0}} = \sqrt{\frac{2 \cdot 10^{-7} \cdot 5.27 \cdot 10^6}{10^{-3}}} = 32.47 \frac{\text{m}}{\text{s}}$$

5) Esercizio 6



$$Q = \rho \cdot \frac{4\pi R^3}{3}$$



$$Q(r) = \rho \cdot \frac{4\pi r^3}{3}$$

$$U_{el} = \frac{1}{2} \int_V V dQ$$

$$\bullet V(r < R) = \frac{\rho}{3\epsilon_0} \cdot \frac{3R^2 - r^2}{2} = \frac{Q}{8\pi\epsilon_0 R} \left(3 - \frac{r^2}{R^2} \right)$$

$$\bullet dQ = \rho \cdot 4\pi r^2 \cdot dr$$

$$U_{el} = \frac{1}{2} \int_0^R \frac{Q}{28\pi\epsilon_0 R} \left(3 - \frac{r^2}{R^2} \right) \rho 4\pi r^2 dr =$$

$$= \frac{Q\rho}{4\epsilon_0 R^3} \int_0^R (3R^2 - r^2) r^2 dr$$

$$= \frac{Q\rho}{4\epsilon_0 R^3} \left\{ 3R^2 \left[\frac{r^3}{3} \right]_0^R - \left[\frac{r^5}{5} \right]_0^R \right\}$$

$$= \frac{Q\rho}{4\epsilon_0 R^3} \cdot \frac{4}{5} R^2 R^3 = \frac{3Q^2}{4\pi\epsilon_0 5R} = \underline{\underline{\frac{3Q^2}{20\pi\epsilon_0 R}}}$$

$$Q \cdot \frac{3Q}{4\pi\epsilon_0 R^3} \cdot \frac{R^2}{5}$$

(b) • carica gusci : $dQ = \rho \cdot 4\pi r^2 dr$

• $V(r, r') = \frac{Q(r)}{4\pi\epsilon_0 r'}$

$$dW = V(r, r') \cdot dQ$$

$$dU = \frac{Q(r)}{4\pi\epsilon_0 r} \cdot dQ = \frac{\rho \cdot \frac{4}{3}\pi r^3}{4\pi\epsilon_0 r} \cdot \rho 4\pi r^2 dr$$

$$= \frac{4}{3} \pi \frac{r^4}{\epsilon_0} \rho^2 dr$$

$$U = \int_0^R \frac{4}{3} \pi \frac{r^4}{\epsilon_0} \rho^2 dr = \frac{4\pi}{3\epsilon_0} \rho^2 \cdot \frac{R^5}{5} = \frac{3Q^2}{4\pi\epsilon_0 5R}$$

U_{el}' ? mode (a)

$$V(r=R) = \frac{\rho R^2}{3\epsilon_0} = \frac{Q}{4\pi\epsilon_0 R}$$

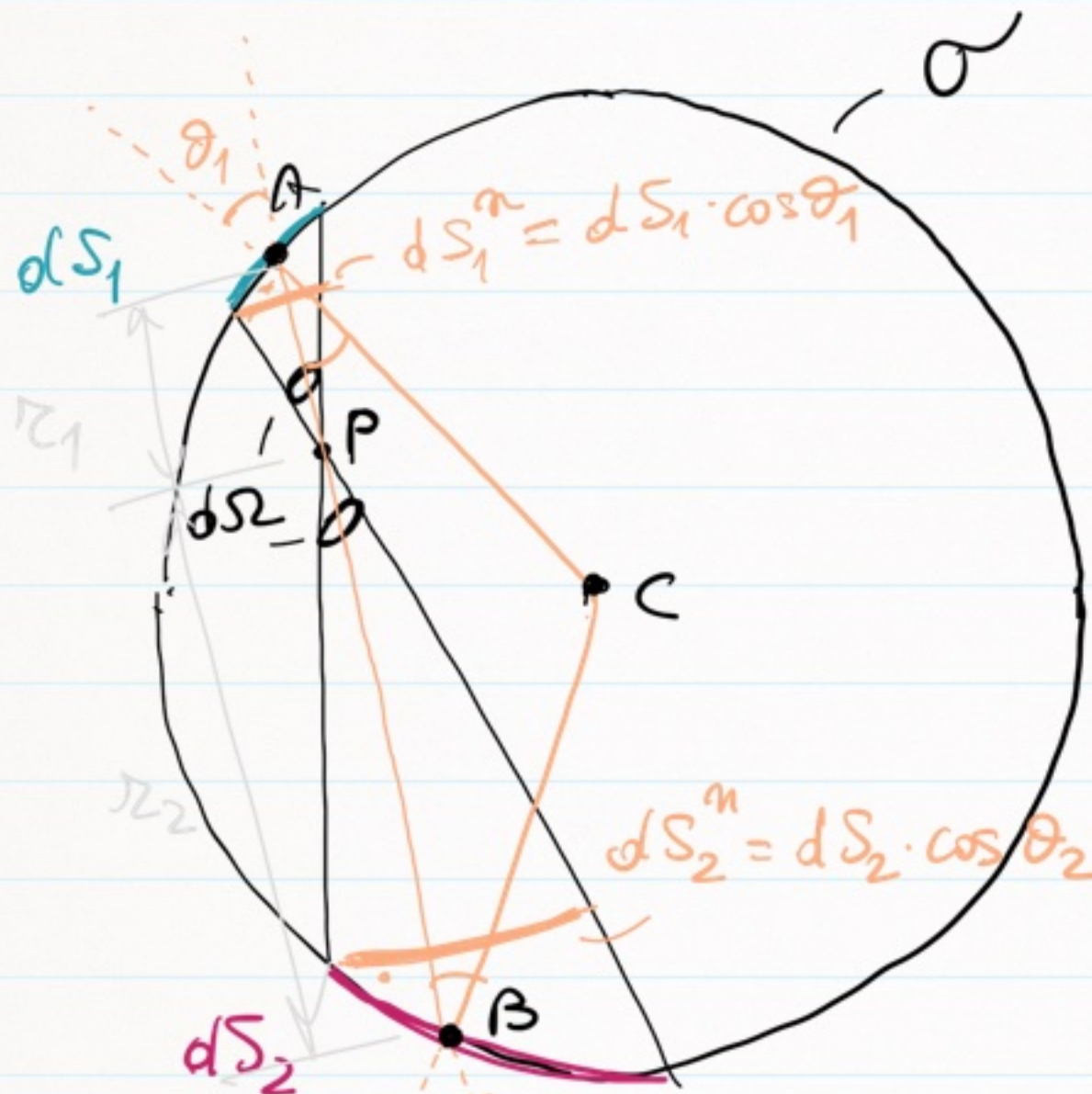
$$U_{el}' = \frac{1}{2} \int_0^R \frac{Q}{4\pi\epsilon_0 R} \cdot \rho 4\pi R^2 dR =$$

$$= \frac{Q\rho}{2\epsilon_0 R} \cdot \frac{R^3}{3} = \frac{Q^2}{8\pi\epsilon_0 R}$$

$$\Delta U_{el} = U_{el}' - U_{el} = \frac{Q^2}{8\pi\epsilon_0 R} - \frac{3Q^2}{20\pi\epsilon_0 R} =$$

$$= - \frac{Q^2}{40\pi\epsilon_0 R} < 0 (!)$$

Soluzione Esercizio 2bis 30/09/20 -



Nel punto P considero due coni di angolo solido al vertice $d\Omega$ ($d\Omega_1 = d\Omega_2 = d\Omega$) che intercettano 2 superfici infinitesime dS_1 e $dS_2 \dots$

$$d\Omega_1 = \frac{dS_1^n}{r_1^2}, \quad d\Omega_2 = \frac{dS_2^n}{r_2^2}, \quad d\Omega_1 = d\Omega_2 = d\Omega$$

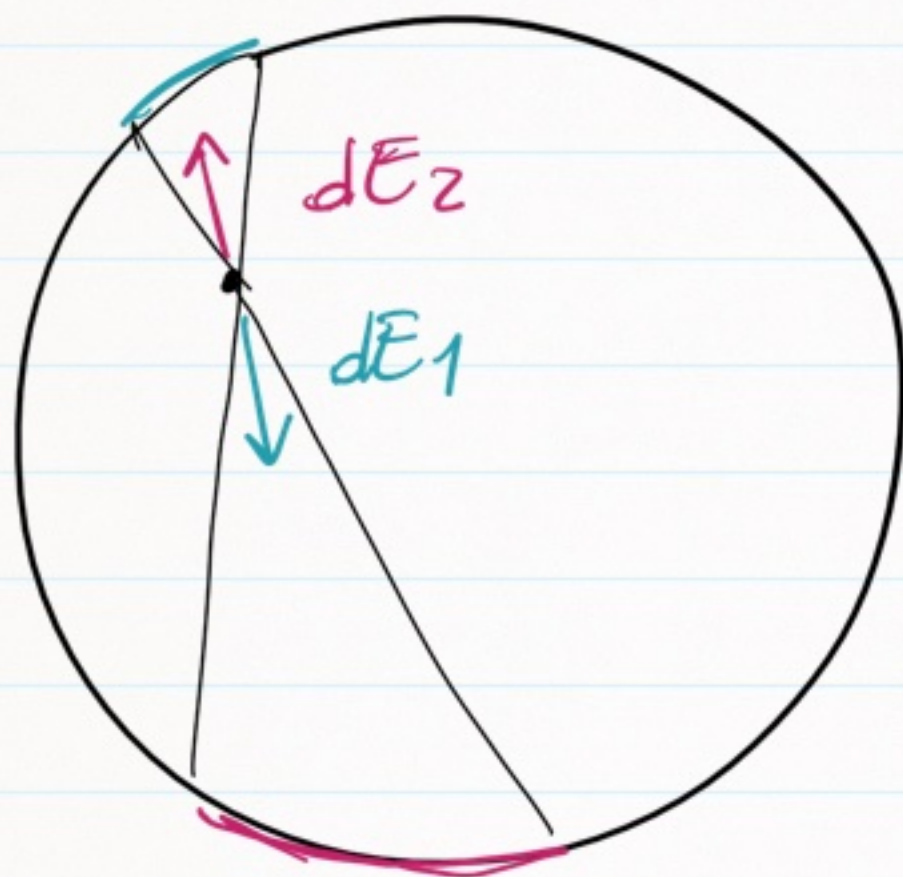
$$dE_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma dS_1}{r_1^2} = \frac{1}{4\pi\epsilon_0} \sigma \frac{dS_1^n}{\cos\theta_1} \cdot \frac{1}{r_1^2} =$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma}{\cos\theta_1} \cdot \frac{dS_1^n}{r_1^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma}{\cos\theta_1} \cdot d\Omega$$

$$dE_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma dS_2}{r_2^2} = \frac{1}{4\pi\epsilon_0} \sigma \frac{dS_2^n}{\cos\theta_2} \cdot \frac{1}{r_2^2} =$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma}{\cos\theta_2} \cdot \frac{dS_2^n}{r_2^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma}{\cos\theta_2} \cdot d\Omega$$

$\cos\theta_1 = \cos\theta_2$ perché $\theta_1 = \theta_2$ (triangolo
isoscele)
ABC



$$dE_1 = dE_2$$

ma opposti !

Quindi, se integro su tutta la sfera, sommo contributi che sono a coppie uguali ed opposti

Risultato : il campo in un generico punto P interno è NULLO